# Mail Server vs AWS SES vs AWS WorkMail

### 1. Introduction

In modern monitoring systems like **Grafana**, alert notifications are a critical component. These alerts are usually delivered through email systems.

There are multiple ways to send emails:

- **Self-hosted Mail Server (SMTP using Postfix)**
- **AWS Simple Email Service (SES)**
- **AWS WorkMail (Enterprise Email Service)**

Each method has its own architecture, use cases, and advantages.

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## 2. Self-Hosted Mail Server (SMTP – Postfix)

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### 2.1 What is a Mail Server?

A mail server is a system that sends and receives emails using protocols like:

- SMTP (Simple Mail Transfer Protocol) → Sending emails
- IMAP / POP3 → Receiving emails

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### 2.2 Components of Mail Server

| Component | Role |
|---|---|
| SMTP Server | Sends emails |
| Mail Transfer Agent (MTA) | Routes emails (Postfix) |
| Mail Delivery Agent (MDA) | Delivers emails |
| DNS (MX Records) | Routes email to server |

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### 2.3 How It Works

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Grafana → SMTP (Postfix) → Recipient Mail Server → Inbox
```

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### 2.4 Key Features

- Full control over email system
- Custom configuration
- Local SMTP relay
- No dependency on external provider

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### 2.5 Security Considerations

- TLS/SSL encryption
- SMTP authentication
- Spam filtering (optional)
- Firewall rules

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### 2.6 Challenges

- Complex setup
- Spam classification issues
- Requires DNS setup:
  - SPF
  - DKIM
  - DMARC
- Maintenance overhead

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### 2.7 Use in Your Task

- Grafana sends alerts using SMTP
- Postfix acts as email relay
- Used for **internal testing and learning**

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## 3. AWS SES (Simple Email Service)

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### 3.1 What is SES?

AWS SES is a **cloud-based email sending service** used for:

- Transactional emails

- Notifications
- Alerts

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### 3.2 Core Features

- Highly scalable
- Pay-as-you-go
- High deliverability
- Supports SMTP and API

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### 3.3 How It Works

```
Grafana → SNS → SES → Email Recipient
```

OR

```
Grafana → SES SMTP → Email
```

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### 3.4 Key Components

| Component | Role |
|---|---|
| Verified Identity | Domain/email verification |
| SMTP Credentials | For sending emails |
| SNS Integration | Alert triggering |
| Email Endpoint | Recipient |

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### 3.5 Security & Compliance

- Domain verification
- DKIM signing
- SPF records
- IAM-based access control

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### 3.6 Deliverability

- High inbox success rate

- Built-in spam protection
- Reputation management

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### 3.7 Limitations

- Sandbox mode (initial restriction)
- Requires verification before sending
- Limited customization compared to self-hosted

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### 3.8 Use in Your Task

- Grafana alerts → SNS → SES → Email
- Reliable alert delivery
- Used for **production-like setup**

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## 4. AWS WorkMail

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### 4.1 What is WorkMail?

AWS WorkMail is a **fully managed business email service** similar to:

- Gmail (Google Workspace)
- Microsoft Outlook

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### 4.2 Core Features

- Managed email service
- Mailbox + calendar
- Secure enterprise communication
- Integration with AWS services

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### 4.3 How It Works

```
Grafana → SNS → WorkMail → User Mailbox
```

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### 4.4 Key Components

| Component | Role |
| --- | --- |
| Organization | Email domain setup |
| Mailbox | User email account |
| SNS Topic | Alert trigger |
| Email Subscription | Delivery |

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### 4.5 Security Features

- Encryption at rest and transit
- IAM integration
- Access control policies

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### 4.6 Limitations

- Cost per mailbox
- Less flexible than SES
- Not ideal for bulk emails

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### 4.7 Use in Your Task

- Grafana alerts → SNS → WorkMail inbox
- Used for **enterprise-level alert delivery testing**

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## 5. Comparison: Mail Server vs SES vs WorkMail

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### Comparison Table

| Feature | Mail Server (Postfix) | AWS SES | AWS WorkMail |
| --- | --- | --- | --- |
| Type | Self-hosted | Cloud service | Managed email |
| Setup | Complex | Moderate | Easy |
| Maintenance | High | Low | Very low |
| Scalability | Limited | High | Medium |

| | | | |
|---|---|---|---|
| Cost | Low (infra cost) | Pay-as-you-go | Per user |
| Deliverability | Low–Medium | High | High |
| Security | Manual setup | Built-in | Built-in |
| Integration | Direct SMTP | SMTP/API/SNS | SNS-based |
| Use Case | Learning/testing | Alerts/notifications | Enterprise email |

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### Simple Explanation

👉 Mail Server = Full control but complex
👉 SES = Best for scalable alerting
👉 WorkMail = Best for enterprise email usage

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## 6. Key Concepts (Important for Your Task)

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### SMTP (Simple Mail Transfer Protocol)

- Used to send emails
- Used by Grafana to send alerts

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### SNS (Simple Notification Service)

- Sends notifications to multiple endpoints
- Acts as middle layer

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### Email Deliverability

- Probability that email reaches inbox
- Depends on:
  - SPF
  - DKIM
  - Reputation

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### Domain Verification

Required in SES to:

- Prevent spam
- Ensure authenticity

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### Alert Pipeline

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Grafana Alert → SMTP / SNS → Email Service → Inbox
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## 7. Why Multiple Email Providers Are Used

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### Benefits

- Compare performance
- Test reliability
- Understand real-world architectures
- Evaluate cost vs performance

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## 8. Conclusion

- Self-hosted mail server gives full control but requires maintenance
- AWS SES provides scalable and reliable email delivery
- AWS WorkMail offers enterprise-level email management

Together, these services help build a **robust alerting system for Grafana**.

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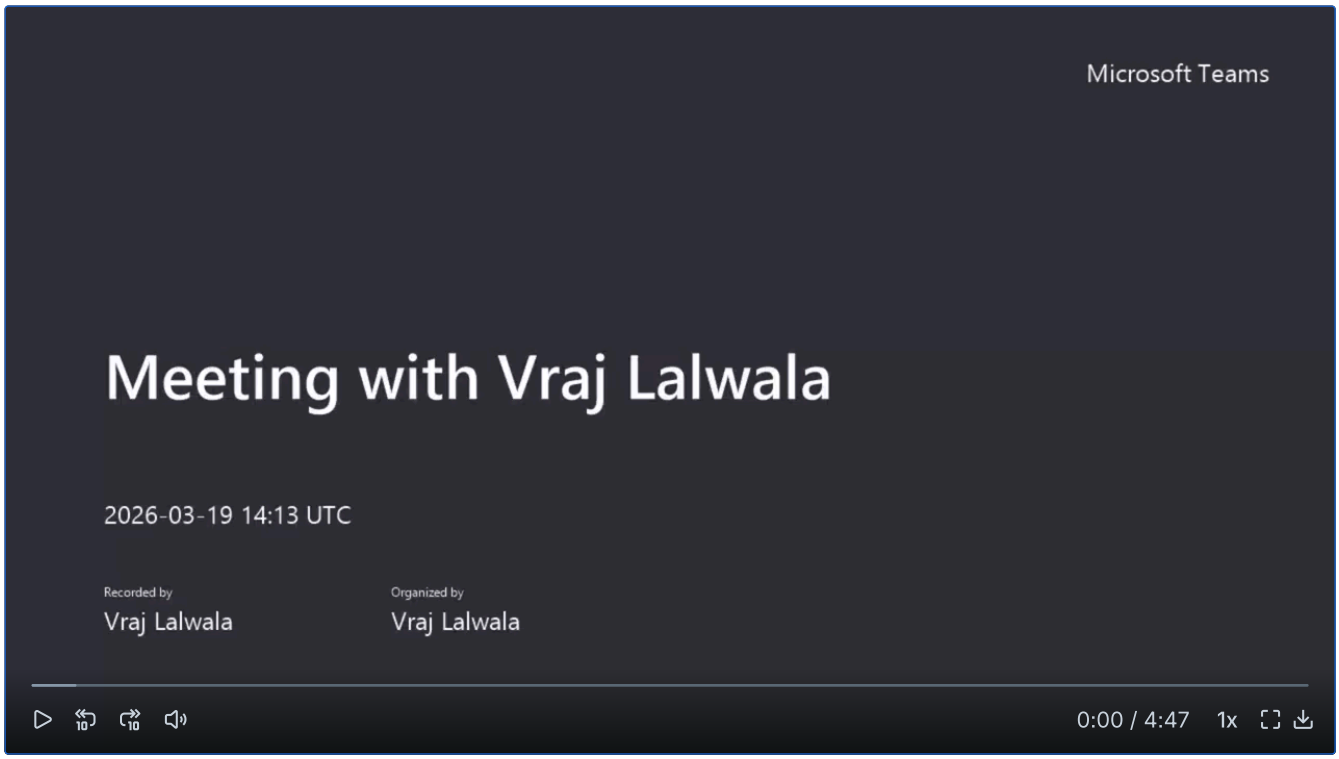
## 9. Video Explanation